PDK Installation and Tool Integration Manual

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Table of Contents

1 Introduction 1
2 Purpose of This Manual 2
3 Basic Concepts 2
3.1.1 What is a PDK? 2
3.1.2 What is Integration? 2
3.1.3 What is PDK_ROOT? 3
4 Pre-Installation Checks 3
5 PDK Installation 3
6 Integrating SKY130 with Magic 4
6.1 Purpose 4
6.2 Pre-Installation 4
6.3 Integration 4
6.4 Post-Installation 4
7 Initial xschem Project Setup for SKY130 4
7.1 Purpose 4
7.2 Pre-Installation 5
7.3 Setup 5
7.4 Post-Installation 5
8 ngspice Integration with SKY130 PDK 5

1 Introduction

This document provides the procedure for installing the Process Design Kit (PDK) and integrating it with the already installed open-source EDA tools.

This manual must be used only after successfully completing the **OpenSource Tools – Installation Manual**. Before starting this document, students must ensure that the required tools such as **xschem**, **ngspice**, and **Magic** are already installed and can be launched from the terminal.

The purpose of this manual is to help students set up a consistent working environment in which the PDK, schematic tool, simulator, and layout tool are correctly linked.

This document focuses on:

- PDK installation
- environment variable setup
- integration of the PDK with xschem
- integration of the PDK with ngspice
- integration of the PDK with Magic
- basic setup verification

2 Purpose of This Manual

The purpose of this manual is to establish the technology environment required for open-source VLSI design practice.

While the tool installation manual covers only the base EDA tools, this document covers the next stage, where the technology files, device models, libraries, and related configuration files are installed and connected to those tools.

3 Basic Concepts

3.1.1 What is a PDK?

A **Process Design Kit (PDK)** is the technology package required for circuit design in a specific semiconductor process. It typically includes:

- device models
- schematic symbols
- layout technology files
- design rules
- parameter files
- simulation support files

3.1.2 What is Integration?

Integration means configuring the installed tools so that all of them point to the same PDK files consistently.

For example:

- xschem should be able to access the PDK symbol libraries
- ngspice should be able to access the device model files

- Magic should be able to access the correct technology file

3.1.3 What is PDK_ROOT?

PDK_ROOT is the environment variable that points to the main directory where the PDK is installed. This variable is commonly used by open-source VLSI tools and flows to locate the required technology files.

4 Pre-Installation Checks

Before starting the PDK installation, ensure the following:

- the base tool installation has been completed
- xschem, ngspice, and Magic are launching correctly
- a stable internet connection is available
- sufficient free disk space is available for PDK files
- the user has sudo privileges
- the user is comfortable editing ~/.bashrc
- the terminal environment is working correctly

5 PDK Installation

Run the following commands to install **open_pdks** with SKY130 support:

```
cd ~/eda_tools/
git clone git://opencircuitdesign.com/open_pdks
cd open_pdks
./configure --enable-sky130-pdk
sudo make
sudo make install
cd ..
```

Note: If no `--prefix` option is provided during configuration, the PDK is typically installed under:

`/usr/local/share/pdk`

Students are advised to run the above steps one at a time and ensure that no errors are reported before proceeding to the next command.

After installation, verify that the SKY130 PDK files are available in the installed PDK location.

A commonly used install location is:

/usr/local/share/pdk/sky130A

You may verify using:

```
ls /usr/local/share/pdk
ls /usr/local/share/pdk/sky130A
```

If the required SKY130 directories are visible, the PDK installation is complete.

6 Integrating SKY130 with Magic

6.1 Purpose

This step links the installed SKY130 technology files to **Magic** so that Magic can open and use the SKY130 technology correctly.

6.2 Pre-Installation

Before proceeding, ensure that:

- Magic has already been installed successfully
- the SKY130 PDK installation is complete
- the SKY130 Magic technology files are available under the installed PDK directory

6.3 Integration

Run the following command:

```
sudo ln -s /usr/local/share/pdk/sky130A/libs.tech/magic/* /usr/local/lib/magic/sys
```

This creates symbolic links so that Magic can access the SKY130 technology support files.

6.4 Post-Installation

You can launch Magic with the SKY130 technology using:

```
magic -T sky130A
```

If Magic opens without error using the SKY130 technology, the integration is successful.

7 Initial xschem Project Setup for SKY130

7.1 Purpose

This step creates a simple project workspace and links the xschem configuration to the installed SKY130 xschem support files.

This helps xschem load the correct PDK-specific symbol and library configuration for SKY130-based work.

7.2 Pre-Installation

Before proceeding, ensure that:

- xschem is already installed
- SKY130 PDK installation is complete
- the SKY130 xschem support files are available inside the installed PDK directory

7.3 Setup

Create a project directory and add a local xschemrc file as follows:

```
cd
mkdir projects
cd projects
mkdir test_xschem_sky130
cd test_xschem_sky130
echo 'source /usr/local/share/pdk/sky130A/libs.tech/xschem/xschemrc' > ./xschemrc
xschem
```

This creates a local project-level xschem configuration file and points it to the SKY130 xschem setup.

7.4 Post-Installation

If xschem launches successfully from this project folder and loads the SKY130 environment correctly, the integration is working.

Important note: Ensure that the SKY130 path used in the xschemrc file matches the installed PDK directory structure on the system.

8 ngspice Integration with SKY130 PDK

Unlike Magic and xschem, ngspice integration is typically completed at the **simulation deck level**.
The required SKY130 model files are referenced within the SPICE netlist using appropriate .lib or .include statements.
Therefore, no separate standalone integration step is usually required during base PDK setup, other than ensuring that the PDK is installed correctly and the model file paths are accessible.
Detailed usage of these model references will be covered in the simulation manual.